PALEOMAGNETIC RESULTS FROM THE CARBONIFEROUS OF NOVA SCOTIA

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Abstract. Throughout the Carboniferous, North America and the southern continents of Gondwana were in collision. New paleomagnetic results from Nova Scotia are in agreement with other determinations from Maritime Canada which suggest that much of the eastern seaboard acted as a collage of microplates, bounded by strike-slip faults, moving in response to changes in the relative plate motion between North America and Gondwana. Paleomagnetic poles from the Early Carboniferous Horton (32 N., 136 E.) and Windsor (36 N., 137 E.) groups, and the Late Carboniferous Morien group (40 N., 131 E.), indicate that during the mid-Carboniferous, Nova Scotia moved northwards out of the dry subtropics and into the equatorial rainy belt. Though Maritime Canada, as a whole, moved northward with respect to the craton along a zone of left-lateral faults (Norumbega-Fredericton faults?), systematic southeasterly declinations in south-central Nova Scotia suggest that a small amount of right-lateral motion may have taken place along the Cobequid-Chedabucto faults.

Introduction

During the Carboniferous and into the Early Permian, the southern continents of Gondwana collided with North America and Europe to form the western half of the supercontinent of Pangea. Evidence of this plate tectonic event is recorded in the Hercynian orogeny of western Europe (Burrett, 1972; Riding, 1974; Badham and Halls, 1975; Johnson, 1978), the Mauritanide orogeny and related Paleozoic deformation in northwest Africa, (Sougy, 1969; Mattauer et al., 1972; Lecorche and Sougy, 1978; Michard and Pique, 1980), and the Alleghanian Orogeny along the Atlantic seaboard of North America (Hatcher, 1972, 1978, 1981). Subduction related intrusives and syntectonic granitic melts, together with broad foreland fold and thrust belts, mark the zone of collision. Associated with these compressional belts is widespread evidence of large-scale movement along extensive strike-slip faults in Europe, North America, and northwest Africa (Lefort and Haworth, 1978; Lefort and Van der Voo, 1981).

The last phases of Hercynian deformation produced dominantly right-lateral offsets throughout western Europe (Arthaud and Matte, 1977; Matte and Burg, 1981). Several dextral strike-slip faults, some with minimum offsets of 100 - 150 km have been mapped, which: 1) strike westward from the northern Pyrenees into the southern end of the Bay of Biscay (Pyrenean fault zone), 2) bisect Brittany (S. Amorican shear zone), and 3) connect subsurface features from the North Sea, southeast into Bohemia (Elbe fracture zone) (Arthaud and Matte, 1977). Additional right-lateral movement may have occurred along reactivated basement trends such as the Tornquist Line and Donetz aulacogen (Ziegler, 1982). The formation of the horst and graben structures of north-central Great Britain may also have been associated with Carboniferous strike-slip tectonics (Dewey, 1982).

The eastern seaboard of North America is cross-cut by numerous NE-SW trending faults (St. Julien and Beland, 1982). The most prominent lineations are the Dover-Hermitage, Reach-Cape Ray, and Cabot faults in Newfoundland; the Hollow, Cobequid-Chedabucto, and Norumbega-Fredericton faults in Maritime Canada; and the Clinton-Newbury, Bloody-Bluff-Lake Char faults in eastern Massachusetts (Figure 1).

The timing, sense, and amount of offset along these, and related, faults is not well known. In only a few cases are offset structures clearly recognized. This difficulty is due in part to the fact that several of these faults follow the trends of mid-Paleozoic sutures (e.g., Cape Ray - Reach faults, McKerrow and Cocks, 1977; Fredericton - Norumbega faults, McKerrow and Ziegler, 1971). Terranes on either side of these faults were literally once oceans apart; moreover,

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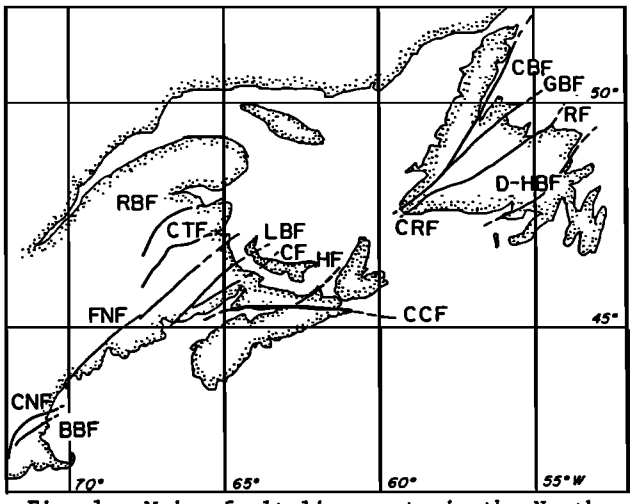


Fig. 1. Major fault lineaments in the Northern Appalachians. BBF=Bloody Bluff Fault, CBF=Cabot Fault, CCF=Cobequid-Chedabucto Fault, CF=Clover Hill-Pollet River Fault, CNF=Clinton-Newbury Fault, CRF=Cape Ray Fault, CTF=Catamaran Fault, D-HBF=Dover-Hermitage Bay Fault, FNF=Fredericton-Norumbega Fault, GBF=Green Bay Fault, HF=Hollow Fault, LBF=Lubec-Belleisle Fault, RBF=Rocky Brook Fault, RF=Reach Fault (modified after: Webb, 1963, 1969; McKerrow and Cocks, 1977; Bradley, 1982).

because these boundaries are fundamental zones of crustal weakness there has been a tendency for reactivation and continued movement (Ruitenberg and McCutcheon, 1982; Keppie, 1982).

The most recent phase of motion along many of these faults appears to have produced small, right-lateral displacements (Bradley, 1982). Dextral offsets of: 30 km along the Taylors Brook fault, Newfoundland (Lock, 1969); 25 km across the Green Bay fault, Newfoundland (Marten, 1971); 100? km along the Long Range fault, Newfoundland (Webb, 1969; Belt, 1969); 16 km along the Catamaran fault, New Brunswick (Anderson, 1972); 9 km along the Clinton-Newbury fault, Massachusetts (Shride, 1976); and 65 km along the Lubec-Belle Isle fault, New Brunswick (Webb, 1963, 1969) have been documented.

If earlier offsets along these faults were sufficiently large (>1000 km), or if individual terranes, caught within these shear zones, have been rotated (>20°), then paleomagnetism can be used to estimate the timing, sense, and magnitude of these displacements. In order to better constrain the timing and movement of Maritime Canada during the Late Paleozoic, paleomagnetic samples were collected from earliest Carboniferous red beds and conglomerates (Horton group), Early Carboniferous red beds and limestones (Windsor group), Middle Carboniferous red beds (Canso and Riverdale groups), and Late Carboniferous siltstones and limestone concretions (Morien group) of Nova Scotia and Cape Breton. Sampling sites were located both north and south of the Cobequid-Chedabucto and Hollow faults, in order to determine if local rotations had taken place.

Previous Studies

The study of the paleomagnetism of Permo-Carboniferous red beds from Maritime Canada has played an important role in both the formulation and testing of tectonic models of the northern Appalachians. One of the earliest studies (Nairn et al., 1959) suggested that the difference in the position of Early Carboniferous paleomagnetic poles from Newfoundland (Codroy fm.), when compared with Carboniferous paleomagnetic determinations from cratonic North America (Barnett formation, Howell and Martinez, 1957; Naco formation, Runcorn, 1956), might be due to a 20° counterclockwise rotation of Newfoundland with respect to North America. To test this hypothesis, paleomagnetic data were collected from a variety of Carboniferous through earliest Permian red bed formations in Quebec, New Brunswick, Nova Scotia, and Prince Edward Island (Dubois, 1959; Black, 1964; Roy, 1963). These results indicated that paleomagnetic directions from Maritime Canada were in agreement with those from Newfoundland; thus, no rotation between these two areas was required. However, the discrepancy between paleomagnetic poles from Maritime Canada and Newfoundland, on one hand, and those from cratonic North America, on the other, still remained.

Through the late 1960's, mostly as a result of the work of Roy and his colleagues, the paleomagnetic history of western Maritime Canada began to emerge. 15° of apparent polar wander (APW) during the Carboniferous was documented by careful demagnetization techniques, and through the use of fold tests which verified that the remanence recorded in the rocks was acquired at, or shortly after the time of deposition (Roy, 1966; Roy and Robertson, 1968; Roy et al., 1968; Roy, 1969; Roy and Park, 1969).

At about the same time, an Early Carboniferous paleomagnetic pole from the Central Appalachians was published (Knowles and Opdyke, 1968). The then available results confirmed that Early Carboniferous paleomagnetic poles from Maritime Canada were displaced 15 - 20 degrees with respect to poles from cratonic North America, but that by Late Carboniferous - earliest Permian times the results from both areas were nearly coincident.

The advent of the cryogenic magnetometer in the mid-1970's permitted the reevaluation of earlier work whose accuracy had been limited by low intensities of remanent magnetization . This new technology also permitted the paleomagnetic study of lithologies, such as carbonates, which previously were considered to be unsuitable. These new data, when viewed in a plate tectonic framework, suggested that the apparent offset of Early Carboniferous paleomagnetic poles may have been the result of micro-plate movements associated with the collison and suturing of Gondwana and the northern continents during the late Paleozoic (Kent and Opdyke, 1979).

Regional Geological Setting

During the Early-Middle Devonian the Acadian island arc terrane collided with the eastern margin of North America, welding Nova Scotia and Cape Breton onto the continent. This plate tec-

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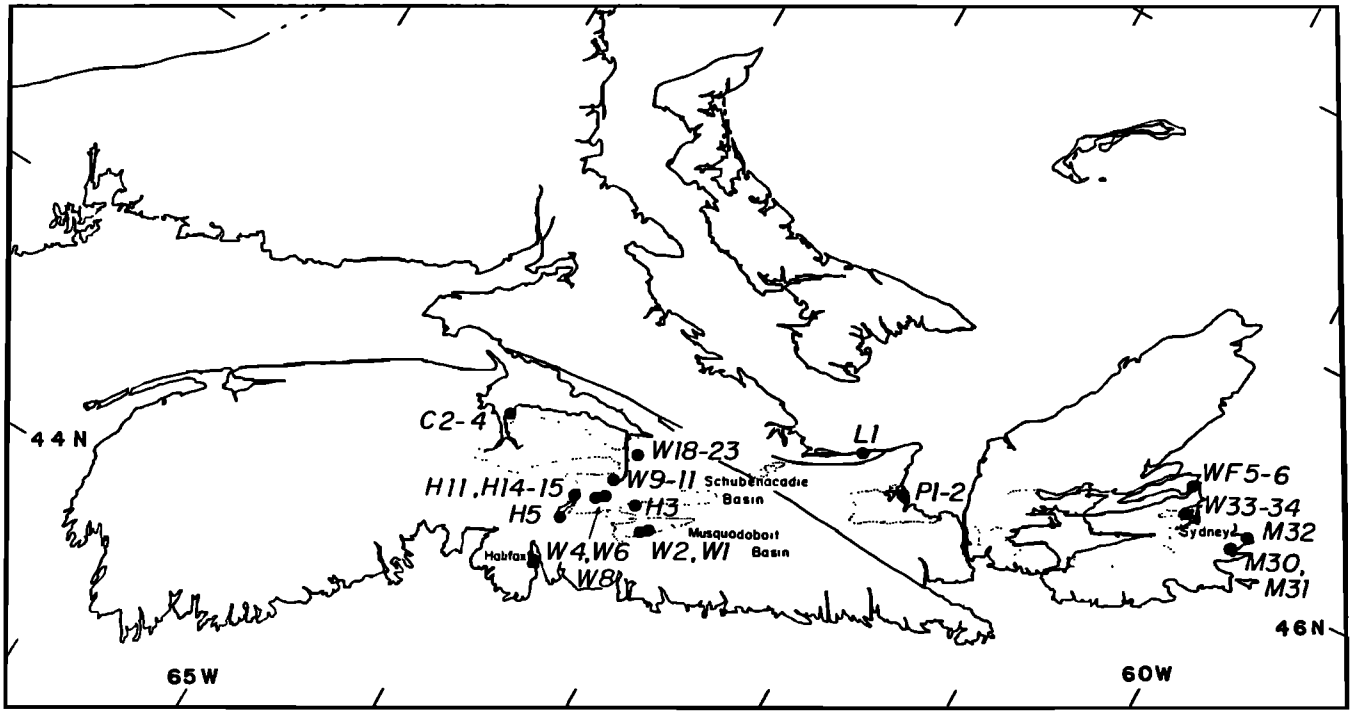


Fig. 2. Location of sites. Dotted line = outline of Carboniferous basins (see Tables I - IV for site identification).

tonic activity represented the final phases of the closure of the Iapetus ocean, which separated the ancient North American continent from northern Europe and England (McKerrow and Ziegler, 1972; McKerrow, 1983). Earlier, during the Late Silurian, closure of a northern extension of Iapetus resulted in Caledonian mountain building in England, Scotland, eastern Greenland, Scandinavia, and Spitzbergen. With the addition of the Acadian terranes the formation of the 'Old Red ' supercontinent was complete.

Evidence of the Acadian orogeny is seen in widespread plutonic igneous activity (Dewey and Kidd, 1974; Loiselle and Ayuso, 1980), which followed the uplift, folding, and penetrative deformation of Paleozoic shelf sequences (Donohoe and Pajarie, 1973). In Nova Scotia and Cape Breton, an angular unconformity separates pre-Acadian shelf and deep water sediments (Arisaig and Meguma groups) from Carboniferous fluviatile and basinal facies.

During the Late Devonian and Early Carboniferous, the tectonic regime changed from compressional to extensional, and a new set of sedimentary basins formed (Howie and Barss, 1975b). The Carboniferous succession in these basins begins with a red, conglomeratic, sandstone and shale sequence, - the Horton group, which is locally over 2000 meters thick. The conglomerates and red beds of the Horton group were deposited in intermontane fault-bound pull-apart basins that opened onto the sea.

As these basins subsided, marine waters transgressed and evaporites accumulated, forming thick deposits of salt and gypsum (Lower Windsor group). The advance and retreat of the seas during the Early Carboniferous is recorded by alternating sequences of red beds, limestones, and gypsum (Giles and Boehner, 1979). The subsequent evaporitic and red bed facies (Upper Windsor and Canso groups) crop out in a wide belt that stretches from eastern New Brunswick, across Nova Scotia and Cape Breton, and into western Newfoundland (Poole et al., 1976).

By the Late Carboniferous the collision of Gondwana and the northern continents was well underway. The intermittent marine facies of the Early Carboniferous were replaced by the dominantly terrestial sediments of the Riverdale, Cumberland, and Pictou groups. Over 4,000 meters of sandstone, shale, and conglomerate, predominantly of fluvial origin continued to accummulate in these pull-apart basins. Economically productive coals were formed at this time suggesting that by the Late Carboniferous, Maritime Canada had moved northwards, out of the dry subtropics, into the wet equatorial rainy belt.

Sampling

144 oriented drill cores were collected from 32 sites in the Horton, Windsor, Canso, Riverdale, and Morien (=Pictou) groups of Nova Scotia and Cape Breton. Samples were obtained both north and south of the Cobequid, Chedabucto, and Hollow faults (Figure 2). The Early Carboniferous conglomeratic red beds of the Horton group,

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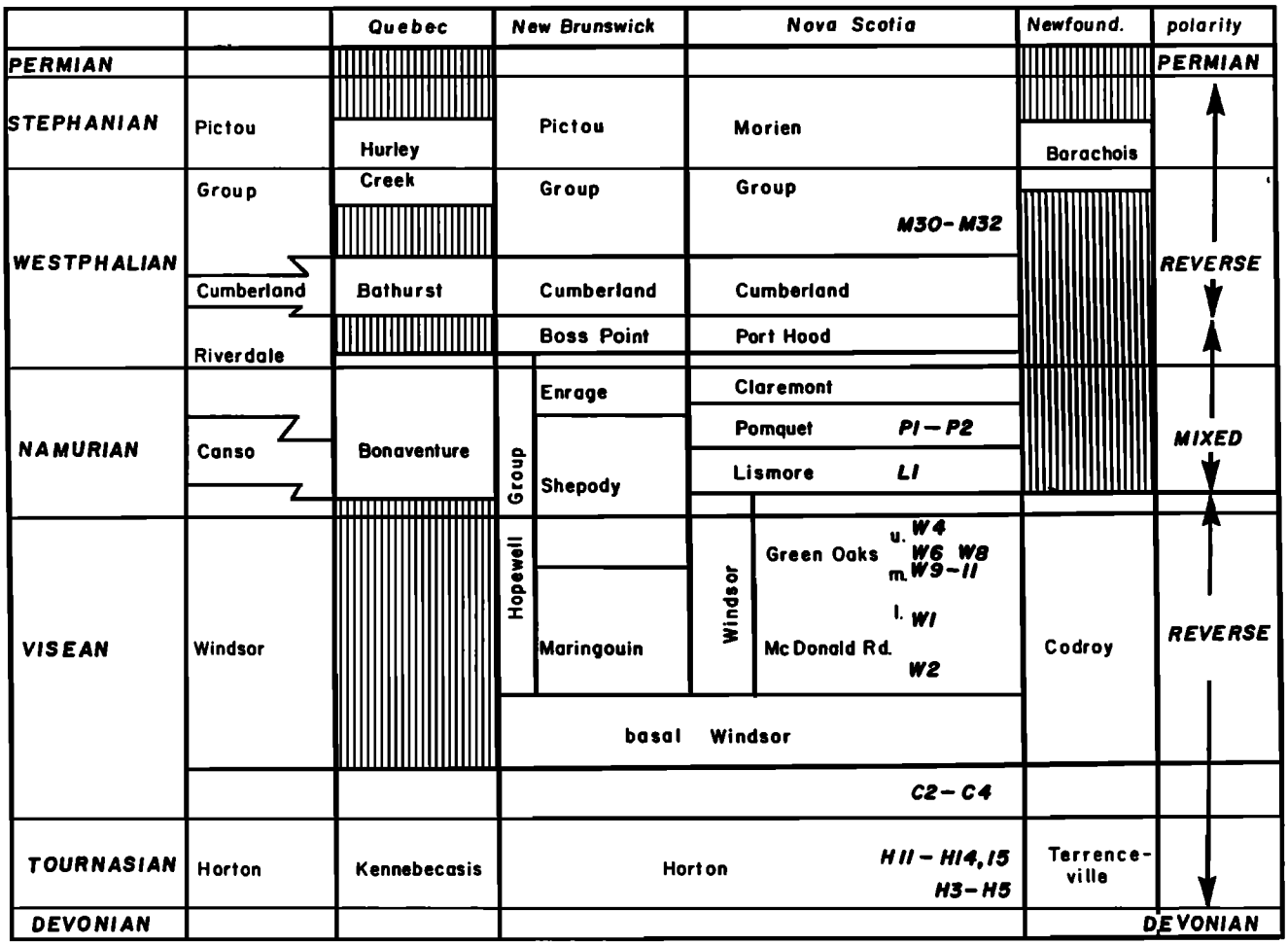


Fig. 3. Stratigraphic relationships (modified after Poole et al., 1976; Howie and Barss, 1975a,b) with sampling localities indicated. Sites WSF5-6, W18-23 are late Visean in age.

and the alternating red beds and limestones of the Windsor group were the main focus of this study.

These units crop out in the Schubenacadie and Musquodoboit basins 50 kilometers N-NE of Halifax. Additional sites in the Windsor were sampled southeast of North Sydney, Cape Breton Island. 6 sites were collected in the Middle Carboniferous Riverdale and Canso groups, and the Late Carboniferous Pictou group. All of these Middle and Late Carboniferous sites lie north of the Cobequid-Chedabucto fault zone (Figure 2). The stratigraphic relationships between sampling sites are illustrated in Figure 3.

Laboratory Technique

The magnetization of the samples was measured using the Superconducting Technology (ScT) magnetometer at the University of Michigan. Because a wide range of lithologies was sampled, a variety of demagnetization techniques were employed to isolate the characteristic direction of magnetization. The red beds of the Horton and Windsor groups were demagnetized in a stepwise fashion using either thermal or chemical demagnetization techniques. The carbonates of the Windsor group were demagnetized using thermal and alternating field (AF) treatments. The siltstones, sandstones, and limestone concretions of the Morien group, as well as the red beds of the Canso and Riverdale groups, were thermally demagnetized.

Thermal demagnetizations were done using a Schonstedt TSD-1 model oven. 10 - 12 samples were heated for not less than 50 minutes, and then were allowed to cool in a field free space. Chemical demagnetizations were performed in a reduced field space, using 8 N HCL as a leaching agent. In order to maximize the surface area on which the acid could act, deep vertical notches were cut into the sample. Between treatment steps, all samples were kept in an area of reduced field strength.

The magnetization remaining after each treatment was plotted using the demagnetization diagrams described by Zijderveld (1967). Characteristic directions of magnetization were determined

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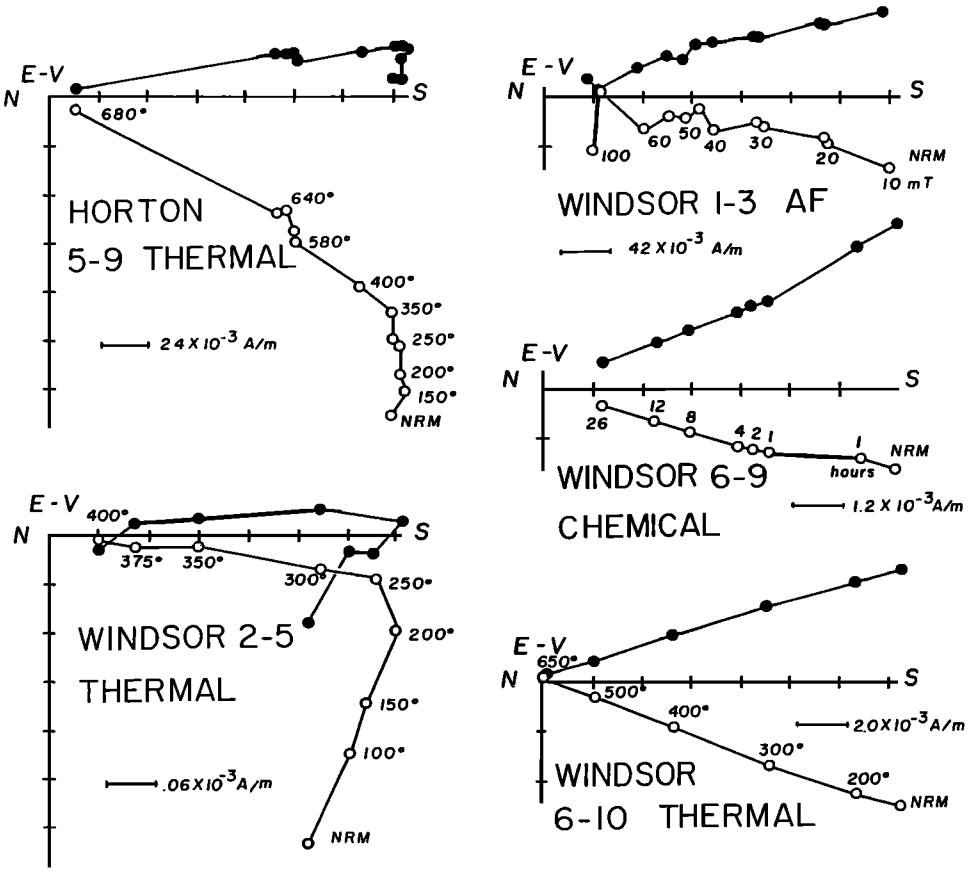


Fig. 4. Thermal demagnetization diagrams for Horton red bed (site H5-#9), Windsor limestones (sites W1-#3, W2-#5), and Windsor red beds (sites W6-#9, #10) plotted in in-situ coordinates: solid circles represent vector endpoints projected onto horizontal plane; open circles, endpoints projected onto N-S vertical plane.

in most cases by vector subtraction along straight line demagnetization trajectories defined by three or more points.

Horton Group

Of the 8 sites collected in the Horton Group, 5 are from a basal conglomeratic unit which unconformably overlies the metagraywackes and schists of the Cambro-Ordovician Meguma group. At one locality (site H5), near Schubenacadie Lake, fist-sized quartzite cobbles were sampled along with a fine-grained hematitic sandstone matrix. The remaining samples were collected from the uppermost part of the Horton group (Cheverie formation), which is exposed in a coastal section one mile southwest of Johnson Cove (Shea et al., 1974). These reddish brown to chocolate fine-grained sandstones and siltstones are disconformably overlain by the Macumber limestone, the basal member of the Windsor group.

The natural remanent magnetization (NRM) of samples from both the upper and lowermost portions of the Horton group is typically southerly directed and shallowly inclined. During thermal demagnetization, the direction shallowed slightly as a steeply downward dipping direction was removed (Figure 4). This initial component, presumably a viscous remanence induced by the present-day magnetic field, was eliminated with thermal treatments between 100° C and 300°C. Between 300° C and 680° C the remanent magnetization decayed linearly to the origin. Samples from the basal part of the Horton group exhibited their highest blocking temperatures near the Curie temperature of hematite (685° C), whereas the magnetization in the red beds from the Cheverie formation was eliminated between 570° C and 620° C.

The site mean characteristic directions from the Horton group are listed in Table 1, and are plotted in Figure 5. Site mean directions steepen slightly after bedding corrections have been applied.

Windsor Group of central Nova Scotia

The limestones and evaporites of the Windsor group record the earliest known Carboniferous

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TABLE 1. Horton Group

ID	LITHOLOGY	n1/n2	Site LAT/LONG	In Situ Direction D/I	k	a95	Bedding Direction D/I	S/D	COMMENTS
H3	conglomerate	6/6	45.1/-63.3	154/+29	22	14.4	-	-	basal Horton
H5	conglomerate	6/6	44.9/-63.6	169/+20	79	7.5	174/+19	168/14	conglom. test
C2	red bed	1/2	45.1/-64.2	166/-18	-	-	164/-10	290/11	Cheverie fm.
C3	red bed	1/1	45.1/-64.2	170/+23	-	-	167/+31	297/10	Cheverie fm.
C4	red bed	2/2	45.1/-64.2	158/+11	62	32.3	156/+17	297/10	Cheverie fm.
H11	red bed	2/2	45.0/-63.6	119/+52	36	43.0	116/+34	017/18	basal Horton
H14	red bed	2/2	45.0/-63.6	162/+12	170	19.3	160/+19	296/10	basal Horton
H15	red bed	6/6	45.0/-63.6	162/+16	184	5.0	160/+23	296/10	basal Horton
					k	a95		k a95	
Formation Mean:		6/8 (reject C2, H11)		162/+19	82	7.5	163/+22	87 8.2	

n1 = samples used; n2 = samples collected; k = precision parameter, a95 = confidence interval, S/D = site mean strike and dip.

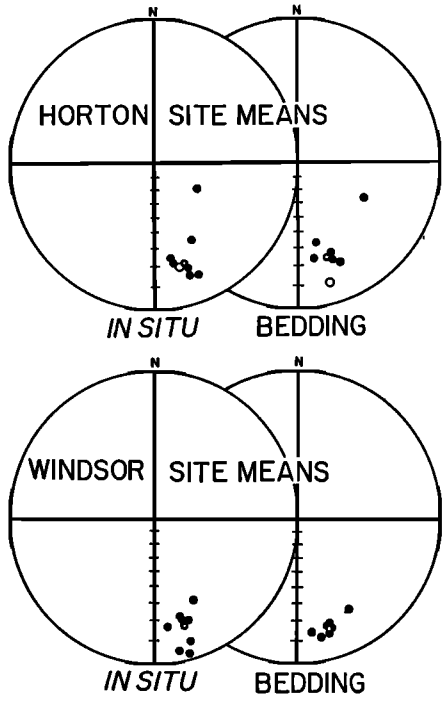


Fig. 5. Horton and Windsor Group characteristic directions (site means) plotted in stereographic projection before and after tilt correction: star = mean of site means. Solid circles indicate positive (downward) inclinations; open circles, negative (upward) inclinations.

marine transgression in Maritime Canada. In the Schubenacadie and Musquodoboit basins of central Nova Scotia, the lower part of the Windsor group consists primarily of gypsum and salt (Carrolls Corner and Stewiacke formations). These evaporites are succeeded by a series of rhythmically alternating beds of red and green siltstone, limestone, and gypsum (MacDonald Road and Green Oaks formations; Giles and Boehner, 1979).

Over 24 transgressive/regressive cycles have been recognized within these two formations. These cycles may reflect local, episodic tectonic events, or they may record global eustatic changes in sea level related to changes in the volume of the Gondwana ice sheet. The approximate period of these cycles (3.0 X 10^5 years), is roughly equivalent to the duration of interglacial epochs during the Pleistocene (Frakes, 1979).

36 of 42 samples collected from the Windsor group, in this area, were obtained from the interbedded red siltstones and dark gray limestone members of the Green Oaks formation. An additional site was obtained from the Ryan Brook member of the MacDonald Road formation (site 2). The uppermost member of the Green Oaks formation, the Kennetcook limestone, is by definition the top of the Windsor group in the Schubenacadie and Musquodoboit basins. It is succeeded conformably by a sequence of terrigenous sediments and evaporites (Canso group) which ,in turn, is overlain

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by a buff, continentally derived sandstone, of Westphalian C-D age (Howie and Barss, 1975a).

Red Beds

The thermal demagnetization diagrams of the Windsor red beds resemble those of the Horton group (Figure 4). The in-situ NRM direction is southerly directed, with a shallow positive inclination. During progressive stepwise thermal demagnetization, this direction tended to shallow slightly; in several samples there was evidence of a small viscous component which was approximately coincident with the present-day magnetic field direction. In all cases, this viscous remanence was eliminated by thermal treatments below 200° C. Between 200° C and 570°C, the characteristic magnetization decayed linearly to the origin. This magnetization was completely eliminated between 620° C and 650° C (Figure 4).

A similar direction of magnetization was revealed during chemical demagnetization (Figure 4). During successive treatments in 8 N HCl, the characteristic magnetization decayed linearly to the origin. After 26 hours of chemical leaching, the strength of the remanent magnetization was reduced to less than 20% of its original intensity.

Limestones

The limestones of the Windsor group were demagnetized using stepwise thermal and alternating field techniques. The NRM directions, though also southerly directed, are 15° to 20° more steeply inclined than the NRM directions observed in the Windsor red beds. This difference in NRM direction is entirely due to the larger component of viscous magnetization retained in the carbonates. This viscous component has a relatively low blocking temperature and was completely removed at 250° C (Figure 4). The remaining component of magnetization decayed linearly to the origin and was eliminated at about 450° C. Again, this southerly, shallow inclination is characteristic of Late Paleozoic directions from Maritime Canada.

Though similar characteristic directions were isolated using stepwise alternating field demagnetization techniques, an anhysteretic component of magnetization was occasionally acquired during AF treatment. This effect was minimized by reversing the orientation of the sample in the demagnetizer, repeating the treatment, and averaging the results (Figure 4). In most cases, 80% of the initial remanent magnetization was removed with AF treatments up to 100 milli-Tesla.

The site mean directions from the Windsor group are listed in Table 2, and are plotted in Figure 5. Site mean directions tend to cluster more tightly after bedding corrections were applied.

Windsor Group of Cape Breton Island

Three additional sites in the Windsor group were drilled on Cape Breton Island, north of the Cobequid - Chedabucto fault zone. Two of these sites were located in, and near, a quarry, 0.5 km east of the town of Pt. Edward. At these localities, samples were collected from alternating beds of gray and red medium- to coarse-grained , cross-bedded sandstones. A third site was sampled by REJ in the Windsor limestones exposed near Shawfield Pt., on the northwestern shores of St. Andrews Channel. The paleomagnetic behaviour of these limestones is similar to that observed in the carbonates of the Schubenacadie and Musquodoboit basins, and the results from these folded beds will be discussed in more detail in the following section.

In a preliminary analysis, the directions in the sandstones from the two sites near Pt. Edward appear to be scattered. This is due in part to the multicomponent nature of the remanence. During thermal demagnetization a shallow, southerly component was removed between 300° C and 500°C. The direction that remained at higher temperatures was northerly directed and inclined steeply downward. The characteristic directions from the three sites on Cape Breton Island tend to be slightly more southerly directed than those directions obtained from sites south of the Cobequid-Chedabucto faults. From this one might conclude that there may have been some movement along this shear zone during the Carboniferous.

Canso and Riverdale Groups

By the mid-Carboniferous (Namurian - early Westphalian), the intermittent marine deposits of the Windsor group were replaced by the fluvial red beds and lacustrine shales of the Canso and Riverdale groups. These units are best exposed north of the Cobequid - Chedabucto fault system, in the Moncton, Cumberland, and Antigonish basins (Figure 2). A preliminary analysis of 13 samples from 2 sites in the Pomquet formation (Riverdale group; Neves and Belt, 1970), and one site in the Lismore formation (Canso group), was made in order to compare paleomagnetic results north of the fault with directions observed in the Schubenacadie and Musquodoboit basins.

Unlike the directions of remanent magnetization from the Windsor and Horton groups which are essentially univectorial, the magnetic signal from the Middle Carboniferous red beds of the Pomquet and Lismore formations is clearly a combination of several components of magnetization. As in the case of the Hopewell, Maringouin, and Riverdale sediments (Roy and Robertson, 1968; Roy and Park, 1969; Roy,1977), progressive thermal demagnetization revealed two nearly anti-parallel directions (Figure 6). Magnetizations in samples from the Pomquet formation exhibit a low blocking temperture component which is directed to the south, and which dips shallowly downward (component A). This direction was progressively removed with thermal treatments between 300° C and 640° C; all that remained at higher temperatures was a component of magnetization with N-NW declinations and shallow, negative inclinations (component B) (Figure 6). This high blocking temperature direction (B) is approximately anti-parallel to the low blocking temperature direction (A).

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TABLE 2. Windsor Group (Flat Lying Sites)

ID	LITHOLOGY	n1/n2	Site LAT/LONG	In Situ Direction D/I	k	a95	Bedding Direction D/I	S/D	COMMENTS
			A. Schubenacadie and Musquodoboit Basins (Central Nova Scotia)						
W1	limestone	7/8	45.0/-63.2	163/+08	184	4.5	164/+12	208/05	l. Green Oaks
W2	limestone	5/5	45.0/-63.2	169/+05	87	8.2	168/+11	295/07	McDonald Road
W4	limestone	4/4	45.0/-63.5	165/+03	77	10.6	-	-	u. Green Oaks
W6	red bed	6/6	45.0/-63.4	161/+18	552	2.8	162/+16	123/04	u. Green Oaks
W8	limestone	5/5	45.0/-63.4	173/+17	142	6.4	172/+14	022/06	u. Green Oaks
W9	limestone	6/6	45.1/-63.4	165/+21	52	9.4	163/+15	028/08	m. Green Oaks
W10	red bed	5/5	45.1/-63.4	164/+19	103	7.6	162/+13	028/08	m. Green Oaks
W11	limestone	3/3	45.1/-63.4	154/+27	72	14.7	149/+20	004/14	m. Green Oaks
					k	a95		k a95	
Formation Mean:		8/8		164/+15	66	6.9	163/+14	118 5.5	
				B. Cape Breton					
W33	sandstone	5/7	46.2/-60.3	158/+02	10	25.6	158/+03	332/06	
W34	sandstone	3/3	46.2/-60.3	170/+00	13	35.6	169/+09	298/11	
					k	a95		k a95	
Formation Mean:		8/10		163/+02	12	16.9	164/+06	11 17.6	

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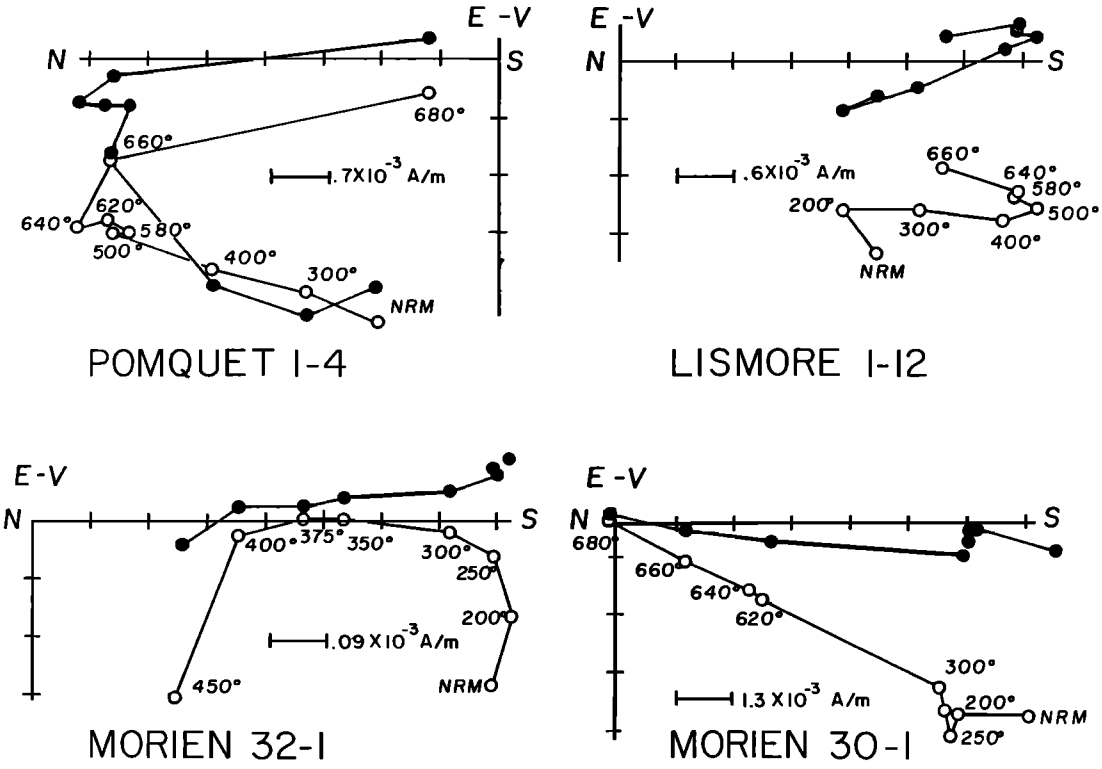


Fig. 6. Pomquet and Lismore formation (sites P1-#4, L1-#12) and Morien group (sites M30-#1, M32-#1) thermal demagnetization diagrams plotted in in situ coordinates. Note dual polarities in samples Pì-#4 and L1-#12. Same plotting conventions as Figure 4.

Similar two-component behaviour is observed in samples from the Lismore formation; however, in this case the low blocking temperature component is northerly directed, and the high blocking temperature component is directed to the south. A northerly and shallowly dipping component of magnetization was removed with thermal treatments between 300° C and 500° C. This direction swung to the south between 500° C and 570° C, then decayed toward the origin with successively higher thermal treatments (Figure 6).

Sample directions for the Pomquet and Lismore formations are combined in Figure 7. Site mean directions (Table 3), in both geographic and bedding coordinates, are similar to the directions found in the older Carboniferous units south of the Cobequid-Chedabucto faults.

Morien Group

Terrestial sedimentation continued into the Late Carboniferous and Early Permian. In the Sydney basin of Cape Breton, the strata correlative with the Late Carboniferous Pictou group of western Maritime Canada are referred to the Morien group (Poole et al., 1976). This unit consists of 1,700 to 2,200 meters of sandstone, conglomerate, siltstone, and shale, as well as, thin, mineable coal seams. Three sites in the Morien were collected: 2 on either side of a gently folded syncline north of the town of Port Morien; and a third along a north-dipping limb of an anticline east of the town of Port Caledonia. A variety of lithologies, including fine grained buff sandstones, calcareous siltstones, and ironstone concretions were collected.

Characteristic directions of magnetization were isolated using stepwise thermal and chemical treatments. In most cases a single component of magnetization was revealed which decayed linearly toward the origin. In a few cases, a steeply dipping downward direction was removed at low temperatures (Figure 6).

The magnetization of samples from site 30 was stable at high temperatures (>500° C). A sizeable fraction of the remanence was eliminated between 570° C and 680° C, indicating that hematite is the primary carrier of the magnetization. In contrast, the magnetization of samples from sites 31 and 32 became unstable above 400° C. Large spurious and often viscous components of magnetization were observed at higher temperatures. These aberrant magnetizations are probably due to the breakdown of unstable iron compounds and the formation of new iron oxides.

The characteristic directions of magnetization for samples from the Morien group are plotted in both geographic and bedding coordinates in Figure 8. Characteristic directions cluster more tightly after the beds are unfolded. The precision parameter, k, increases from 14.4 (k_1) before the tilt correction has been applied, to 31.8 (k_2) after unfolding. The ratio of the pre-

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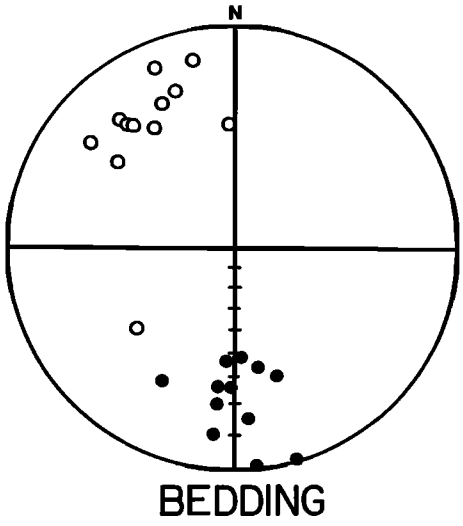


Fig. 7. Pomquet formation characteristic directions (individual sample directions) illustrating dual polarities recorded in each sample (bedding coordinates). Same plotting conventions as Figure 5.

cision parameters (k_1/k_2) indicates that these directions pass the fold test at the 99% confidence level.

The site mean direction for the Morien group after unfolding (D = 171, I =6.5) represents a magnetization which was acquired sometime after deposition (Late Carboniferous), but before folding (earliest Permian). The paleomagnetic pole position calculated on the basis of this direction is in good agreement with Late Carboniferous - Early Permian paleomagnetic south poles from Maritime Canada and cratonic North America (observed, -40.2/-48.9; expected, -43.0/-48.6, Van der Voo et al., this volume). This indicates that there has been little relative movement within Maritime Canada, or between Maritime Canada and cratonic North America since the latest Carboniferous.

Age of Acquisition of Remanence in the Horton and Windsor groups

Conglomerate Test: Horton Group

In order to better define the time of acquisition of the magnetic signature in rocks of the Horton group, a conglomerate test was carried out. Six samples were collected from the basal conglomerate which unconformably overlies the schists of the Meguma group near the south end of Schubenacadie Lake. Three samples were obtained from fist-sized quartzite cobbles, an additional three samples were taken from the surrounding red bed matrix. The characteristic directions from these samples are plotted in Figure 9. The tight cluster of directions indicates that the primary remanence, presumably once carried by the cobbles, has been overprinted with a direction similar to that observed in the matrix.

There are two possible ways to produce this apparent overprint: chemically and thermally. In the first instance, iron, carried in solution from the surrounding matrix, infiltrates the cobbles and new iron oxides grow along grain boundaries and microfractures. Since the iron is derived from the red bed matrix, it is conceivable that the remanence was acquired during deposition. Alternatively, assuming a thermal resetting, both the cobbles and matrix may have acquired a secondary magnetization during the period of uplift and cooling following an orogenic event. In the following section, we will argue that a chemical remagnetization, contemporaneous with deposition, is the more likely choice.

Fold Test: Windsor Group

Throughout most of its outcrop belt the Windsor carbonates and red beds are nearly flat lying; dips usually range between 5 - 14 degrees. However, on Cape Breton Island, near Shawfield Pt. along the northwestern shores of St. Andrews Channel, the Windsor carbonates are gently folded into an east-west trending syncline.

Characteristic directions which are dispersed before unfolding, cluster tightly after the limbs have been rotated to horizontal (Table 4, Figure 10). The precision parameter, k, increases from 35.5 (k_1),to 75.9 (k_2), after unfolding. This result strongly suggests that the remanence was acquired before the Late Carboniferous - Early Permian folding. The site mean direction (D=168°, I=+15°) is similar to the direction observed in flat lying beds from Cape Breton Island and central Nova Scotia.

An additional fold test was attempted at a site along the northwestern edge of the Schubenacadie basin, 1 km east of the town of Green Creek. At that locality, along the stream bed of Andrew Creek, a continuous section of thin bedded, dark gray limestones is contorted into a series of asymmetric folds. This site lies within 20 km of the Cobequid - Chedabucto shear zone and the observed deformation is probably related to strike-slip movement along these faults.

Characteristic directions from these sites are dispersed both before, and after, unfolding (Table 4). Moreover, site mean inclinations are 20 - 25 degrees more negative than inclinations from the stable parts of the Schubenacadie and Musquodoboit basins, suggesting that these directions may not be representative of Windsor directions further to the south. The more negative inclinations may represent an Early Mesozoic remagnetization.

Four arguments may be made in favor of primary magnetizations in the Windsor and Horton groups. (1) The positive fold test in the Windsor limestones from Cape Breton indicates that the remanence was acquired before the Late Carboniferous - Early Permian. Furthermore, the site mean direction is similar to directions observed in flat lying beds from Cape Breton and central Nova Scotia. (2) The site mean directions from the Windsor red beds and carbonates of the Schubenac-

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TABLE 3. Canso, Riverdale, and Morien Groups

ID	LITHOLOGY	n1/n2	Site LAT/LONG	In Situ Direction D/I	k	a95	Bedding Direction D/I	S/D	COMMENTS
				A. Canso and Riverdale Groups					
"A" Component									
P01	red bed	5/5	45.6/-61.7	182/+09	38	12.5	186/+23	228/20	Pomquet fm.
P02	red bed	4/4	45.6/-61.7	190/-17	6	40.4	186/-05	220/23	Pomquet fm.
L01	red bed	4/4	45.7/-62.2	161/+26	68	11.2	172/+31	182/21	Lismore fm.
"B" Component									
P01	red bed	5/5	45.6/-61.7	341/+03	32	12.5	342/-15	228/20	Pomquet fm.
P02	red bed	2/2	45.6/-61.7	316/-02	36	42.7	316/-24	220/23	Pomquet fm.
L01	red bed	4/4	45.7/-62.2	314/-01	129	8.1	316/-16	182/21	Lismore fm.
				B. Morien Group (Fold Test)					
M30	siltstone	7/7	46.1/-59.9	189/+31	202	4.3	175/+02	053/32	
M31	siltstone	7/8	46.1/-59.9	160/-05	23	12.7	160/+07	235/11	
M32	siltstone	7/10	46.2/-59.9	178/+01	60	7.9	178/+11	270/10	
					k	a95		k a95	
Formation Mean:		21/25		176/+09	14	8.7	171/+06	32 5.7	

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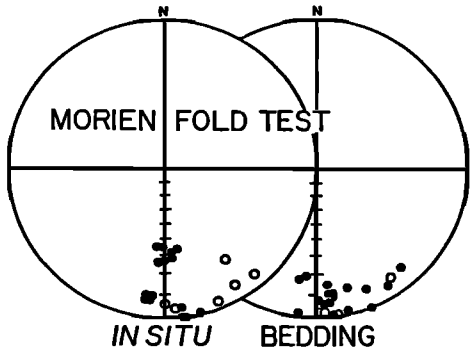


Fig. 8. Morien Group characteristic directions (individual sample directions) plotted before and after tilt correction. k1 (in situ) = 14, k2 (after unfolding) = 32. Same plotting conventions as Figure 5.

adie and Musquodoboit basins cluster more tightly when the tilt corrections are applied (k_1 = 66.1, k_2 = 118.0). (3) The direction of magnetization, and the resulting pole positions are in excellent agreement with previously published results from Early and Middle Carboniferous red beds that also pass the fold test (Figure 11 and 12, see Table 5). (4) The progression of pole positions from the Horton, through the Windsor, and into the Morien, is unidirectional and falls along the expected APW path for Maritime Canada.

Magnetic Stratigraphy

From the results of this study, a preliminary reversal time scale can be constructed for the Carboniferous. Samples were obtained from units representing the Tournaisian, Visean, Namurian, and Westphalian stages. Reversed directions were found in the youngest (Morien) and oldest (Horton and Windsor) groups. The interval between these groups is characterized by a zone of mixed polarities (Canso and Riverdale). The approximate length of these intervals is illustrated in Figure 3.

The reversed polarity sequence of Late Carboniferous age corresponds to the lowermost portion of the Kiaman Interval. This period of reverse polarity extended from the Early Triassic (Scythian), back through the Permian, and into the Late Carboniferous. Though the lower boundary of the Kiaman Interval has been the subject of numerous investigations (e.g., McMahon and Strangway, 1968; Valencio et al., 1977), its lower limit is still poorly known. The polarity sequence observed in mid to Late Carboniferous units from Nova Scotia, when combined with the results of Roy (1966, 1969) and Roy et al. (1968) suggests that the base of the Kiaman Interval lies within the early Westphalian (A-B).

The zone of mixed polarity extends from the earliest Westphalian back to the early Namurian. The presence of dual polarities in the Hopewell Group of eastern New Brunswick (Enrage, Shepody, Maringouin formations), suggests that these units may be entirely Namurian in age (Roy, 1977). A period of reverse polarity resumes in the Visean and extends back through the Early Carboniferous, into the Late Devonian.

Conclusion: Tectonic Implications

There is little doubt that by the Late Carboniferous - Early Permian, Gondwana had collided with North America and Europe to form the nucleus of Pangea. After the final phases of suturing, movement ceased along the various strike-slip faults which had been active during collision. It is not clear, however, to what extent Maritime Canada, during the Early and Middle Carboniferous, acted as a collage of microplates, bounded by strike-slip faults moving in response to changes in the relative plate motion between North America and Gondwana.

Within Maritime Canada there are several strike-slip faults along which some of this motion may have been accommodated (Figure 1). The most prominent lineations are the Cabot, Hollow, Cobequid-Chedabucto, and the Norumbega-Fredericton faults. Estimates of strike-slip motion along these faults have varied from no movement (Ludman, 1981), to a few hundred kilometers (Webb, 1969; McCabe,1981), to an indeterminate amount of strike-slip motion (Wones and Thompson, 1979). The amount of displacement along these faults can be estimated paleomagnetically by comparing the direction of Carboniferous magnetizations on either sides of these faults.

Magnetic declinations for the Early and Late Carboniferous are plotted in Figure 11, with corresponding paleolatitudes indicated. From this figure we see that magnetic declinations both north and south of the major faults zones are similar. South-southeasterly declinations are

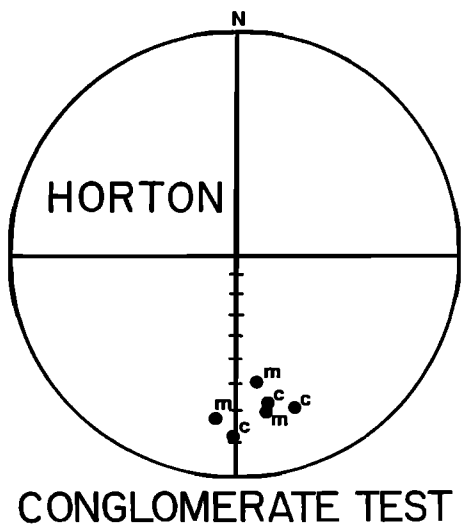


Fig. 9. Horton conglomerate test. Characteristic directions (M = matrix, C = cobble) plotted in in-situ coordinates. Same plotting conventions as Figure 5.

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TABLE 4. Windsor Group (Fold Tests)

ID	LITHOLOGY	n1/n2	Site LAT/LONG	In Situ Direction D/I	k	a95	Bedding Direction D/I	S/D	COMMENTS
				A. Cape Breton (Shawfield Pt.)					
WF5	limestone	3/3	46.3/-60.4	171/+04	123	11.2	173/+15	225/17	
WF6	limestone	5/5	46.3/-60.4	167/+23	60	10.0	166/+14	53/10	
					k	a95		k a95	
Formation Mean:		8/8		169/+16	36	9.4	168/+15	76 6.4	
				B. Schubenacadie Basin					
W18	limestone	3/4	45.3/-63.4	146/+03	823	8.7	156/-57	044/62	
W19	limestone	1/3	45.3/-63.4	141/+18	-	-	146/-62	046/80	
W20	limestone	3/4	45.3/-63.4	136/-36	29	23.1	135/-21	220/15	
W21	limestone	2/2	45.3/-63.4	146/+05	13	76.8	145/+10	288/09	
W22	limestone	3/3	45.3/-63.4	150/-16	35	21.2	151/-07	270/10	
W23	limestone	3/3	45.3/-63.4	142/-02	140	10.5	142/+06	269/10	
					k	a95		k a95	
Formation Mean:		6/6		144/-05	17.7	16.4	145/-21	6.7 27.9	

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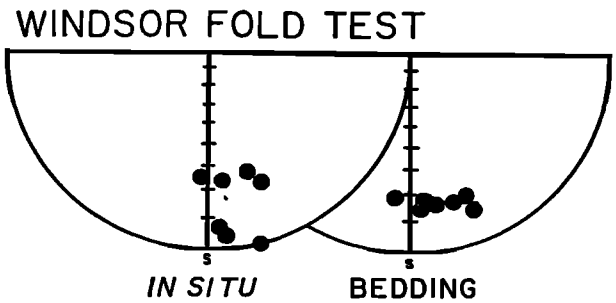


Fig. 10. Windsor fold test, Shawfield Point. Characteristic directions plotted before and after tilt correction. k1 (in situ) = 35.5, k2 (after unfolding) = 75.9. Same plotting conventions as Figure 5.

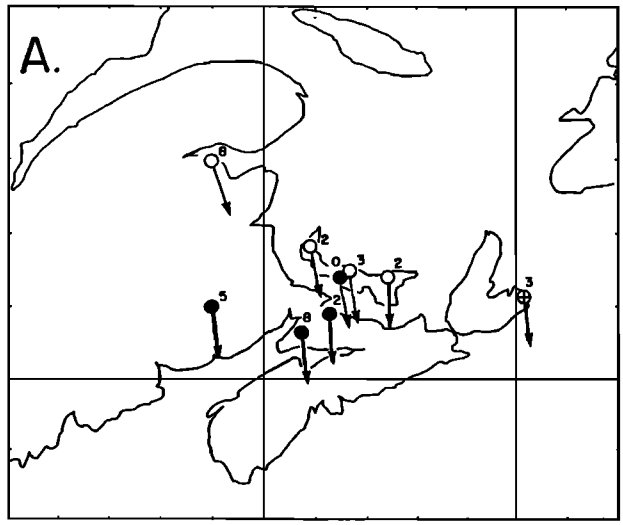
found in eastern Quebec, central and southern New Brunswick, across Nova Scotia, and into western Newfoundland.

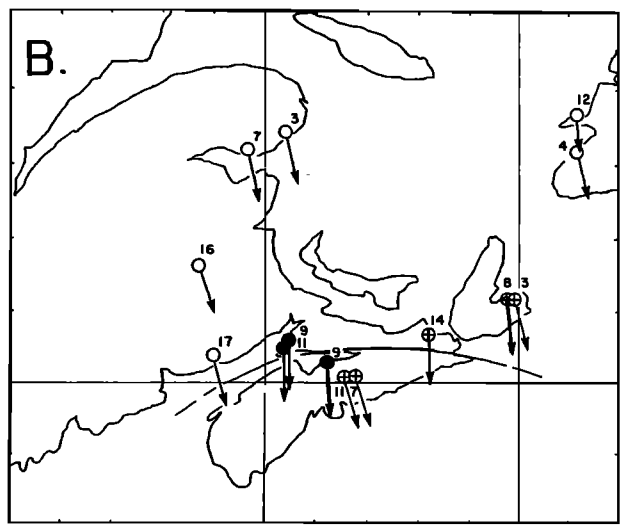
Late Carboniferous paleolatitudes place Maritime Canada on the equator. This latitudinal orientation is supported by the presence of numerous coal seams in the Cumberland and Pictou groups. The magnetic declinations of Late Carboniferous age are nearly identical across Maritime Canada indicating that the motion along the various strike-slip faults had ceased by latest Carboniferous - earliest Permian times.

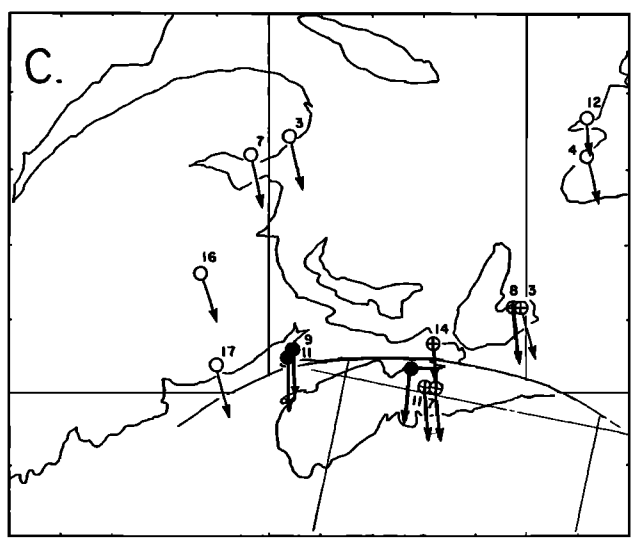
Early Carboniferous magnetic declinations, through less uniformly directed, are also remarkably similar across the width of the belt. In general, declinations are more southeasterly than those of Late Carboniferous age. Early Carboniferous paleomagnetic results place Nova Scotia 10° - 12° farther to the south, closer to the dry subtropics. This position is consistent with the widespread occurrence of gypsum and halite in the Windsor and Canso groups.

It should be noted that several of the magnetic directions plotted in Figure 11 are based on older, less reliable determinations (open circles). These determinations were published before careful demagnetization procedures became routine and are, therefore, suspect. The demagnetization behaviour of red beds in this study suggests that, though the inclinations determined from these earlier studies are possibly aberrant, the observed declinations may be representative of the characteristic directions. During stepwise demagnetization, the direction of magnetizaton changes within a nearly N-S vertical plane; though inclinations shallow, declinations are not severely affected.

A closer look at the Early Carboniferous directions reveals that though the declinations are similar across much of Maritime Canada, south of the Cobequid-Chedabucto faults, the directions seem to be aligned slightly more to the southeast. From stratigraphic and tectonic evidence it is clear that these faults were active during the Carboniferous (Bradley, 1982). Various amounts of offset have been proposed (Webb, 1963, 1969; McCabe, 1981; Keppie, 1982); in each case, right-lateral displacement is required. In Figure 11 south-central Nova Scotia has been rotated







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20°, clockwise, along the trace of the Cobequid-Chedabucto faults. This rotation brings magnetic declinations, north and south of the faults, into slightly better alignment. A more robust test of the tectonic models which require right-lateral strike slip movement across the Cobequid-Chedabucto faults, is presented by Spariosu et al. (this volune). Secondary magnetizations observed in the Siluro-Devonian White Rock formation (south of the Cobequid-Chedabucto faults), are rotated 20 ° in a counterclockwise sense with respect to expected Late Paleozoic declinations from Maritime Canada. The combined paleomagnetic data are consistent with tectonic and geologic evidence which suggests that there has been approximately 150 km of right-lateral offset along the Cobequid-Chedabucto faults during the Carboniferous.

Though internally consistent, the paleomagnetic results from Maritime Canada do not agree with paleomagnetic determinations from cratonic North America. A comparison of Early and Middle Carboniferous poles (Figure 12) reveals that paleomagnetic determinations from these two areas fall into two distinct groups. The separation between the two clusters has been cited as evidence for large scale sinistral strike-slip motion along the eastern seaboard during the Carboniferous (Kent and Opdyke, 1979; Kent, 1982). The minimum amount of movement required to superimpose the two sets of poles is 10°, or about 1000 kilometers.

The results of this study support the interpretation that there was large scale left-lateral strike-slip movement along the eastern seaboard during the Carboniferous. Early Carboniferous paleomagnetic results from the Horton and Windsor groups place Maritime Canada 10 - 15 degrees south of its present position with respect to cratonic North America. This latitudinal difference is nearly eliminated by Late Carboniferous - Early Permian times. The pole position from the Morien Group is in good agreement with paleomagnetic determinations from Maritime Canada and cratonic North America (Table V).

If Maritime Canada was once displaced south of its present position with respect to cratonic North America, where are the faults along which subsequent left-lateral displacement took place? The western limit of strike-slip faulting can be mapped by noting which paleomagnetic determinations from Maritime Canada appear to be displaced with respect to results from craton. Unfortunately, most of the reliable data are from eastern New Brunswick and Nova Scotia, areas which were well east of any shear zone. Though there are determinations from Quebec and western New Brunswick (Bonaventure and Bathurst formations), the results are based solely on NRM directions

Fig. 11. Magnetic declinations and paleolatitudes. Closed circles = reliable results, open circle = unreliable results, circle with cross = results, this paper. A. Late Carboniferous; B. Early-Middle Carboniferous; C. Early Carboniferous with 20 degree clockwise rotation along Cobequid-Chedabucto Fault. All paleolatitudes are southern hemisphere.

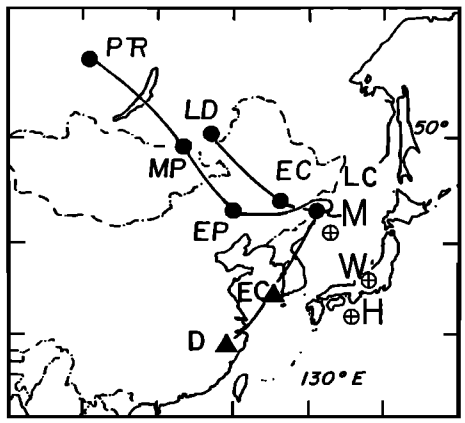


Fig. 12. Late Paleozoic apparent polar wander paths for Maritime Canada (triangles) and cratonic North America (circles). Results for this study are indicated by M. Morien Group; W. Windsor Group; H. Horton Group.

from formations whose ages are uncertain.

The most likely candidate to accommodate the required left-lateral displacement is the Norumbega-Fredericton fault system. However, controversy exists as to whether displacements of 100's of kilometers along the Norumbega-Fredericton fault system are possible (Ludman, 1981; Wones and Thompson, 1979). It should be noted that the amount of displacement required along these faults can be reduced if some of the latitudinal offset can be taken up by crustal shortening and compression, as in the case of Tibet.

This trans-Acadian shear zone must continue northward towards Newfoundland. Preliminary paleomagnetic results from Newfoundland suggest that the fault zone may pass west of the island (Barachois fm., Opdyke, 1981); however, geologic evidence that links the Long Range of Newfoundland with easternmost Labrador contradicts this interpretation (Rodgers, 1970, p. 149). Alternatively, the fault may follow the suture of the Iapetus Ocean (Cape Ray - Reach faults), or slice northwards along the western edge of the Avalon platform (Dover-Hermitage Bay faults). A paleomagnetic study of the Terrenceville formation (Kent, 1982) indicates that the Avalon platform of easternmost Newfoundland is displaced with respect to cratonic North America.

The most important aspect of the paleomagnetic results from Maritime Canada is the regional consistency of Carboniferous directions. In fact, reliable determinations from Maritime Canada outnumber results from cratonic North America by a ratio of 2:1. It is tempting to argue that the apparent difference between the cluster of poles from Maritime Canada and North America is due to poor sampling. Confidence intervals clearly overlap. The apparent difference in paleolatitude between these two areas may be reduced as additional results, especially from the craton, become available.

Though poor sampling is a possible explanation, this interpretation does not take into account

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TABLE 5. Carboniferous Paleomagnetic Poles

Age		Formation	Pole Lat./Long.	Reference
		I. Late Carboniferous		
A. Cratonic North America				
1. 285	LC-EP	Maroon	41/133	Christensen and Helsley, 1974; Christensen, 1974
2. 288	LC	L. Casper	46/129	Diehl and Shive, 1981
3. 290	Missour.	Brush Creek	36/124	Payne et al., 1981
4. 290	Missour.	Buffalo slt.	27/123	Payne et al., 1981
5. 290	LC	pre-Kiaman	37/122	McMahon and Strangway, 1968
6. 290	LC	coal pole	37/132	Kopacz and Noltimier, 1976
7. 295	Desmoin.	Naco ss.	46/133	Runcorn, 1956
8. 295	Desmoin.	Naco ss.	23/130	Collinson and Runcorn, 1960
9. 295	Desmoin.	Mazon Creek	39/133	Scotese, 1981
Mean Pole: n=2 (#2,9)			43/131	K=227.8 A95=16.6
B. Maritime Canada				
1. 280	LC-EP	Prince Edward I.	39/123	Roy, 1963
2. 280	LC-EP	P.E.I. (east)	41/122	Black, 1964
3. 280	LC-EP	P.E.I. (central)	40/128	Black, 1964
4. 280	LC-EP	P.E.I. (north)	40/129	Black, 1964
5. 280	LC-EP	Prince Edward I.	42/133	Roy, 1966
6. 280	LC-EP	Pictou Group	41/132	Roy, 1966
7. 290	LC	Hurley Creek	39/125	Roy et al., 1968
8. 290	LC	Morien Group	40/131	This paper
9. 305	West. A/B	Cumberland Grp.	36/125	Roy, 1969
10. 305	West. A/B	Bathurst	32/135	Black, 1964
Mean Pole: n=5 (#1,6,7,8,9)			40/129	K=457 A95=3.5
		II. Early Carboniferous		
A. Cratonic North America				
1. 320	L. Miss.	Mauch Chunk	43/127	Knowles and Opdyke, 1968
2. 335	E. Miss.	Barnett	49/119	Kent and Opdyke, 1979
3. 338	Osagean	St. Joe Is.	39/132	Scott, 1979
4. ?	Miss.?	'Deadwood'	47/117	Collinson and Runcorn, 1960
Mean Pole: n=3 (#1,2,3)			44/126	K=139.2 A95=10.5
B. Maritime Canada				
1. 310	MC	pre-Pictou	24/133	Black, 1964
2. 310	L. Namur.	'Riverdale'	34/124	Roy, 1977
3. 315	Namur.?	Bonaventure	34/131	Roy, 1966
4. 315	Namur.	Hopewell	34/119	Roy and Park, 1969
5. 315	Namur.	Lismore, Pomquet	29/122	This paper
6. 330	Visean	Codroy	43/139	Nairn et al., 1959
7. 330	Visean	Codroy	30/127	Black, 1964
8. 330	?Visean	Maringouin	34/117	Roy and Robertson, 1968
9. 330	Visean	Windsor	36/137	This paper
10. 340	Tournais.	Kennebecasis	22/133	Black, 1964
11. 340	Tournais.	Horton	32/136	This paper
12. 340	Tournais.	Terrenceville	27/124	Kent, 1982
Mean Pole: n=5 (#2,4,8,9,11)			34/127	K=105 A95=7.5

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the fact that Late Devonian, as well as Carboniferous, paleomagnetic poles from Maritime Canada are displaced with respect to the craton (Van der Voo and Scotese, 1981). A post-Devonian, pre-earliest Permian left-lateral strike-slip boundary between Maritime Canada and cratonic North America is required by the paleomagnetic data.

As we begin to understand the evolution of the Northern Appalachians in terms of plate tectonic processes, a pattern emerges which is similar to that observed in the Cordillera of western North America. This pattern is complete with exotic terranes, sutures, accretionary belts, and large scale strike-slip faults. Unravelling this pattern requires a multi-disciplinary approach which carefully weighs the evidence provided by stratigraphy, tectonics, paleontology, and paleomagnetism. Hopefully, such an approach will eventually lend us the insights necessary to understand the complex plate tectonic development of regions such as the Northern Appalachians.

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